IRCTC: Cloud Scaling for Railway Ticket Booking (Tatkal Booking Spikes)

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1. Sources found

- **Official:** PIB release, Ministry of Railways (29 Jun 2025) on the new Passenger Reservation System; PIB backgrounder "Four Decades of Passenger Reservation" (6 Aug 2026); Lok Sabha Unstarred Question 2787 (6 Aug 2025, Aadhaar-authenticated Tatkal).
- **News quoting officials:** Lok Sabha reply on Tatkal bots (IANS, Dec 2025); Business Today and others on the July 2026 beta and August 2026 migration; RTI on IRCTC convenience fees (Jun 2026); Aadhaar/OTP circular coverage (Jun 2025).
- **Vendor / partner:** CloudThat case study on AWS security services for IRCTC (undated, no figures); Red Hat innovation award to CRIS (2014).
- **Context only:** system-design blogs (hypothetical designs, used only for our own design). Full list with links: docs/sources.md.

2. Case study salient points

- **Who runs what.** The Passenger Reservation System (PRS, since 1986) is built by CRIS. IRCTC is the online front end. From Jun 2025 to Jun 2026, 65.08 crore reserved tickets were booked, 89% online (about 20.5 lakh passengers/day).
- **Problem.** Tatkal opens at 10:00 (AC) and 11:00 (non-AC), a sharp burst far above average load. The legacy PRS dates from 2010 (Itanium servers, OpenVMS) and handles about 32,000 bookings and 4 lakh enquiries per minute.
- **Demand-side fix.** Aadhaar-authenticated Tatkal (1 Jul 2025), OTP (15 Jul 2025), agents blocked for the first 30 minutes, 3.02 crore suspicious IDs deactivated, Akamai anti-bot filtering and WAF.
- **Supply-side fix.** A new "cloud-enabled" PRS by CRIS, designed for ten times the load: over 1.5 lakh bookings/min and over 40 lakh enquiries/min. Beta website live 15 Jul 2026; phased train migration with load testing began Aug 2026.
- **Outcome so far.** Confirmed Tatkal availability time up about 65% on 322 trains (Dec 2025). Early beta: Tatkal bookings completed within 3 minutes up over 5%. The full capacity target is a projection, not yet proven at Tatkal peak.
- **Cost.** No official cost for the new PRS found (Rs 1,000 cr appears in one news report only). Convenience-fee revenue was Rs 954 cr in FY25 (RTI).
- **Evidence gap.** No official source gives the new PRS's internal architecture or cloud provider, so we do not claim IRCTC runs on AWS.
- **Link to our exercise.** The Tatkal burst maps to the assigned exercise: a booking app behind a load balancer with autoscaling, load-tested with Locust/k6, plotting response time against instance count.